

CS6440: Introduction to Health Informatics Spr24

Practicum Sprint #6 Deliverable #3 Academic Paper

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Abstract—We introduce *SkinWise*, a web-based diagnostic tool empowering healthcare providers, including hospitals and physicians, in the timely detection and effective management of skin cancer, with a specific emphasis on melanoma. Leveraging advanced machine learning techniques and datasets like the Skin Cancer ISIC dataset and FHIR, *SkinWise* offers innovative solutions for accurate cancer classification and management.

1 BACKGROUND

Skin cancer poses a significant and growing public health challenge worldwide with its steadily rising across diverse populations. Among its various forms, melanoma stands out for its particularly severe prognosis and elevated mortality rates, underscoring the critical importance of early detection and intervention Khazaei et al., 2019. Traditional approaches to skin cancer diagnosis have relied heavily on visual inspection and dermoscopic analysis of skin lesions (Loescher et al., 2013), often suffering from subjectivity and variability in interpretation, leading to diagnostic inaccuracies and delayed treatment initiation (Ruffano et al., 2018). However, the challenges of skin cancer diagnosis extend beyond mere detection. Effective management of skin cancer cases requires comprehensive solutions that go beyond diagnosis alone (Khan et al., 2022). The complexities of managing skin cancer patients lie in the timely coordination of care, treatment planning, and ongoing monitoring. Addressing these complexities necessitates innovative approaches that not only enhance diagnostic accuracy but also streamline patient management processes (Najjar, 2023).

2 PROBLEM STATEMENT

Skin cancer represents a substantial global health concern, and ensuring effective management of skin cancer cases demands holistic solutions that encompass both diagnosis and ongoing patient care. The intricate nature of managing individuals with skin cancer, encompassing treatment planning and vigilant monitoring, highlights the imperative for advanced methodologies in both detection and

patient management. As a result, there is an urgent demand for standardized, objective diagnostic tools to enhance the precision and availability of skin cancer diagnosis and management. The deployment of an automated skin cancer detection and management system like *SkinWise* is essential in improving patient outcomes by facilitating more efficient and effective management of skin cancer cases on a global scale.

3 SOLUTION

We develop SkinWise, a web-based platform offering an innovative solution to the complex challenges surrounding skin cancer diagnosis. Our platform employs advanced machine learning techniques, specifically a deep learning model trained on the Skin Cancer ISIC dataset, to analyze images of skin lesions and provide accurate diagnostic insights. The development process of SkinWise involved a meticulous approach to model training, validation, and refinement to ensure optimal performance in skin cancer classification. Leveraging the extensive Skin Cancer ISIC Dataset, we trained our deep learning model on a diverse range of skin lesion images, encompassing various skin conditions and manifestations. Additionally, the integration of Fast Healthcare Interoperability Resources (FHIR) standards ensures seamless compatibility with various electronic health record systems, facilitating efficient data exchange and interoperability within the healthcare ecosystem. Rigorous validation procedures were implemented to assess the model's accuracy and robustness, with iterative refinement cycles conducted to enhance performance further.

4 OUTCOME AND FURTHER WORK

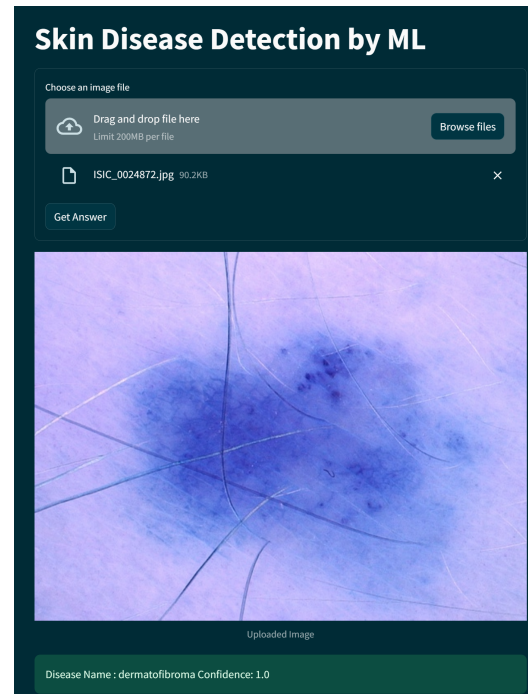
4.1 Outcome

SkinWise offers a user-friendly interface that enables healthcare professionals to upload patient images seamlessly and receive real-time diagnostic suggestions. The platform's intuitive design facilitates ease of use and accessibility, empowering healthcare providers with actionable insights for timely intervention and patient management. Specifically, we've achieved significant milestones throughout the development journey of SkinWise, particularly in enhancing the application's capabilities in both patient management and diagnostic assistance. We successfully integrated key functionalities, including linking our web application to the HAPI FHIR public server (see Figure 1a), a crucial step in ensuring

seamless interoperability within healthcare systems, and an integration of the skin disease classification model (see Figure 1b), which empowered healthcare professionals with real-time diagnostic suggestions, facilitating timely intervention and patient management. Moreover, to ensure cohesion and progress, we have regular group meetings for internal deliberations and strategic planning and use the GitHub repository to maintain updates and version control. We also successfully deployed our Streamlit application on an Amazon Web Services (AWS) Elastic Compute Cloud (EC2) instance utilizing Docker for containerization (see Figure 1a). We made the application accessible via the instance’s public IP and DNS, showcasing a practical application of cloud computing and containerization for deploying Python applications.

The screenshot shows a web application interface for patient data entry. At the top, there are tabs for 'Data Entry' and 'Patients'. The main section is titled 'Data Entry for Patient'. It contains several input fields: 'Select Birth Day' (dropdown), 'Select Birth Month' (dropdown), 'Select Birth Year' (dropdown), 'First Name' (text input), 'Family Name' (text input), 'Diagnosis' (text input), and a 'Save Data' button. Below this, there is a file upload section with a 'Choose an image file' label, a 'Drag and drop file here' area, and a 'Browse files' button. A file named 'ISIC_0028394.jpg' (61.5KB) is shown as uploaded. Further down, there are text inputs for 'Patient's Name', 'Patient's Contact Number', 'Doctor's Name', and 'Doctor's Contact Number', followed by a 'Get Answer' button. At the bottom, a green bar displays the results: 'Patient Name: zxf Doctor Name: zzz Disease Name : melanocytic nevi Confidence: 1.0'.

(a) SkinWise UI with integration of machine learning classification model



(b) Machine learning classification model result

Figure 1—Screenshots of SkinWise UI and machine learning classification model result

4.2 Further Work

As *SkinWise* evolves, our research and development efforts will prioritize enhancing the platform’s capabilities and extending its utility. This entails refining the deep learning model to achieve heightened accuracy and sensitivity in skin cancer detection. Additionally, we aim to explore and implement additional features

to bolster comprehensive patient management and follow-up care. Also, a user study could help to gather feedback and insights from users. Furthermore, ongoing engagement with healthcare providers and stakeholders will guide iterative improvements, ensuring alignment with evolving clinical needs and standards.

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